

Anish Kumar

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Education

DY Patil International University, Akurdi, Pune

Aug 2024 – May 2028

- Bachelor of Engineering (Computer Science and Engineering) | SGPA: 8.85/10.0
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks

Projects

Video Streaming Platform | React · Node.js · FFmpeg · PostgreSQL · Redis · BullMQ · MinIO | [GitHub](#)

Jan 2026

- Built an HLS transcoding pipeline using FFmpeg to generate adaptive 360p/720p streams stored on MinIO (S3-compatible); benchmarked sub-3s stream start time on uploads averaging 500 MB
- Designed an event-driven upload pipeline with presigned S3 URLs, BullMQ background workers (concurrency: 2), and Socket.IO for real-time progress updates — handled 20+ concurrent uploads without queue starvation
- Implemented a hybrid recommendation engine combining trending scores, personalized feeds, collaborative filtering, and co-watch signals; chose a 40/40/20 weighting to balance discovery with relevance for new vs. returning users
- Built full-text search with PostgreSQL tsvector, trigram fuzzy matching (pg_trgm), and Redis-cached ranking; reduced average query response time to under 80 ms on a 10,000-row dataset

LinkHub — Link-in-Bio Platform | React · Node.js · PostgreSQL · Stripe · Docker | [GitHub](#)

Feb 2026

- Built a full-stack link-in-bio platform with a real-time mobile preview (Zustand subscriptions), a 7-section design panel, and a wallpaper engine supporting gradients, image effects, patterns, and video backgrounds
- Engineered a three-tier freemium subscription system (Free/Pro/Premium) with Stripe Checkout, monthly/annual billing, and cancel/resume flows; integrated a ProGate feature-gating component using React.cloneElement to intercept clicks without modifying wrapped child components
- Implemented Google OAuth (Passport.js) + JWT auth with per-user Zustand localStorage persistence rehydrated on every login; built analytics tracking for profile views and per-link click events with geo and device metadata
- Containerised the full stack with Docker Compose (React/Nginx frontend, Node.js backend, PostgreSQL) with long-lived cache headers (Cache-Control: max-age=31536000) and React Router fallback support

Discord Clone — Real-Time Chat & Voice | Node.js · WebSockets · WebRTC | [GitHub](#)

Sep 2025

- Built real-time messaging and voice channels supporting 50+ concurrent WebSocket connections with sub-100 ms message delivery; implemented channel-based message routing with per-channel broadcast isolation
- Implemented peer-to-peer voice communication using WebRTC with ICE negotiation and STUN server integration; added JWT-based authentication, channel management, and message history persistence via PostgreSQL

Technical Skills

Languages: C++, Python, JavaScript, TypeScript

Frontend: React.js, Tailwind CSS, GSAP, Zustand

Backend: Node.js, Express.js, WebSockets, WebRTC, BullMQ, Socket.IO, Passport.js

Database & Storage: PostgreSQL, MongoDB, Prisma ORM, Redis, MinIO (S3-compatible)

Tools & DevOps: Git/GitHub, Docker, FFmpeg, AWS, Stripe

Leadership & Achievements

- **FullStack Lead — Code-Decode Club** (Official Coding Club, DY Patil International University, 2024–Present)
- **Full-Stack Bootcamp Instructor** — Designed and led a 10-day intensive bootcamp for 80+ students covering React, Node.js, PostgreSQL, and JWT authentication; created structured lesson plans, live-coding sessions, and capstone project briefs
- **Hackathon Magnitude 2.0** — Built an AI-powered crop management platform for farmers featuring Google Gemini-based pest detection from images and a real-time irrigation monitoring dashboard; competed against teams from multiple institutions